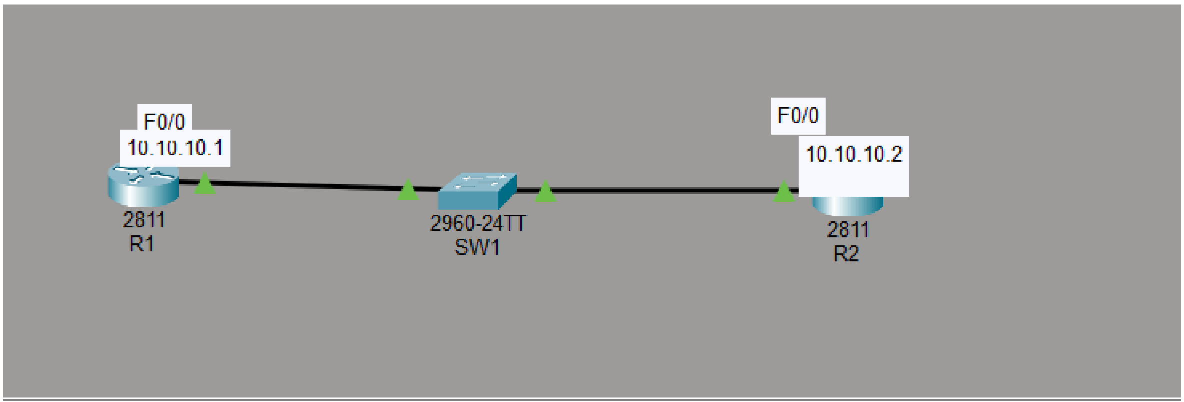
Week 5 Practical

Initial configuration

Screenshot of completed topology



28.Configuration of hostname for Router 1

```
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

29.cofiguration of hostname for Router 2

```
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#
```

30. configuration of hostname for switch 1

```
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#hostname SW1
SW1(config)#
%SYS-5-CONFIG_I: Configured from console by console
```

31.Configuring IP address for R1

```
Router#enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Ethernet 0/0
%Invalid interface type and number
Router(config)#interface Fastethernet 0/0
Router(config-if)#ip address 10.10.10.1
% Incomplete command.
Router(config-if)#ip address 10.10.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

32. Configuring IP address for R2

```
Router#enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 10.10.10.2 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

33.Configuring SW1 IP address

```
SW1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface vlan 1
SW1(config-if)#ip address 10.10.10.10 255.255.255.0
SW1(config-if)#no shutdown
```

34. Pinging SW1 to R2

```
SW1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/3/12 ms
```

CDP configuration

36. Verify the directly attached Cisco neighbours using CDP.

```
R1>enable
R1#show cdp
Global CDP information:
    Sending CDP packets every 60 seconds
    Sending a holdtime value of 180 seconds
    Sending CDPv2 advertisements is enabled
```

37. Prevent R1 from discovering SW1 via CDP : disable CDP on R1's interface facing SW1 specifically ('no cdp enable' on that interface), not globally, since Task 38 below re-enables CDP globally.

```
    Sending CDPv2 advertisements is enabled
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int F0/0
R1(config-if)#no cdp enable
R1(config-if)#exit
R1(config)#show cdp
            ^
% Invalid input detected at '^' marker.

R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
show cdp
Global CDP information:
    Sending CDP packets every 60 seconds
    Sending a holdtime value of 180 seconds
    Sending CDPv2 advertisements is enabled
R1#show cdp neigbhor
                ^
% Invalid input detected at '^' marker.

R1#show cdp neigbhor
                ^
% Invalid input detected at '^' marker.

R1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
R1#
```

38. Flush the CDP cache on R1 with 'no cdp run' then 'cdp run' in global configuration mode.

```
R1>enable
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no cdp
% Incomplete command.
R1(config)#no cdp run
R1(config)#cdp run
```

39. Verify that R1 can no longer see SW1 via CDP (the interface-level disable from Task 37 should still be in effect).

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability  Platform    Port ID
R1#
```

Switch troubleshooting

40. Verify the status of the switch port connected to R2 with 'show ip interface brief' : it should show up/up.

```
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
```

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41. Shut down the interface connected to R2 and check the status again : it should show administratively down/down.

```
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual administratively down down
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
 --More--
```

42. Bring the interface back up and verify its speed and duplex settings.

```
R2>enable
R2#
R2#configure terminal
Enter configuration commands, one per line.  End with
R2(config)#interface FastEthernet0/0
R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#
```

43. Set the duplex on SW1's interface to half (leave R2 as-is) and verify the state of the interface.

```
to up
duplex full
SW1(config-if)#duplex half
SW1(config-if)#show ip interface brief
                 ^
% Invalid input detected at '^' marker.

SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
 --More--
```

44.set duplex

```
SW1#
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface FastEthernet0/2
SW1(config-if)#duplex full
SW1(config-if)#
```

45. Setting speed to 10 mhps

```
SW1#
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface FastEthernet0/2
SW1(config-if)#duplex full
SW1(config-if)#speed 100
SW1(config-if)#speed 10
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
 --More--
```

46. Check whether the interface on R2 is still operational, and note its status


```
SW1#
SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface FastEthernet0/2
SW1(config-if)#duplex full
SW1(config-if)#speed 100
SW1(config-if)#speed 10
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
 --More--
```


The interface for R2 is up and manually configured even with 10mhps , it works well

46. When you set a duplex mismatch between SW1 and R2, what symptoms did you observe, and why does a mismatch cause this behaviour?

When you set a duplex mismatch between SW1 and R2, the link stayed up but performance dropped noticeably. You would have seen slow throughput, packet loss, and a rise in collision or error counters on the interfaces. This happens because one side in full duplex tries to send and receive at the same time, while the half-duplex side only allows one direction at once. The mismatch leads to late collisions and retransmissions, which degrade network efficiency even though the physical connection remains active.